PHINS — AI & BI Optimization: Implementation Summary

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Audience: Investors and technical due-diligence reviewers.

Status: Shipped to the platform (additive, integrity-preserving). See the companion *AI & BI Optimization Review* for the full thesis and roadmap.

This note summarizes the first wave of optimizations turning PHINS's AI/BI story from *interface* into *substance*, implemented so that **data integrity remains flawless**: every new layer is append-only or a read-only derived view, and nothing here moves money or changes a policy/claim state on its own.

What shipped

AI agent upgrades (decisioning becomes auditable and tunable)

1. **Append-only AI decision log.** Every automated decision the AI agent makes — quote, underwriting, claim triage — is now recorded immutably with its inputs, output, score, thresholds, model version, and segment. Human overrides are linked to the original decision *additively* (the original is never rewritten). This is the keystone that enables auditability and learning.
1. **Per-segment thresholds.** Underwriting decision thresholds are now resolved per cohort (age band × occupation) instead of one global cut-off. Defaults are identical to the previous global values, so behavior is unchanged until an operator explicitly promotes calibrated thresholds.
1. **Recommend-only calibration.** A calibration routine reads the decision log and recommends better per-segment thresholds from real human-override outcomes. It only *suggests*; promotion is an explicit, audited action.
1. **Model registry with rules as fallback.** A lightweight registry can serve a transparent, versioned model (the actuarially-accepted GLM/logistic family) *behind* the existing rules. With no model artifact present (today's default), the deterministic rule engine remains fully authoritative — so explainability is preserved and behavior is unchanged.

BI upgrades (faster, consistent, investor-grade)

1. **One canonical KPI definition.** Loss ratio, approval rate, MRR/ARR, net

worth, and receivables ratio now have a single source-of-truth module, so the number on the executive dashboard always matches finance — answering "which number is true?" with one answer.

1 **Activated dashboard cache.** Executive, customer, supplier, and delivery dashboards are cached against a content fingerprint of their inputs. Repeated reads are served instantly; the moment the underlying data changes, the cache recomputes — so a cached dashboard can never contradict live state.

Data-integrity guardrails

1 **Require-database fail-fast (opt-in).** With one environment flag, the platform refuses to start on the volatile in-memory fallback, preventing the silent split-brain where the request path and reporting read different stores. Off by default; turning it on makes the "durable, reconcilable" guarantee enforceable in production.

The integrity contract (why this is safe)

- **Append-only:** decision records and ledger events are never mutated in place; corrections are new, linked entries.
- **Derived layers recommend, never post:** AI scores and BI dashboards/caches are read-only views over canonical state. Money moves only through the existing ledger and accounting engines, under rule or human gates.
- **Best-effort, never fatal:** decision logging and model lookups can never raise into the live decision path — the deterministic rules always answer.

What's next (ready for prioritization)

- Persist the decision log and actuarial simulations to the database as durable, versioned artifacts.
- Hourly precomputed dashboard snapshots and continuous, historized integrity checks.
- Canonical event emission so BI/AI derive from one hash-chained event stream.
- Decimal-money discipline across the remaining financial services.

These are detailed, with file-level specifications and integrity guardrails, in the *AI & BI Optimization Review* (linked from the pitch dashboard).